We know how to take a lot of limits, but some are still annoying / impossible with our current knowledge.

Consider $\lim\limits_{x \to 1} \dfrac{\ln(x)}{x-1}$

this is an $\frac{0}{0}$ case, which is dificult to simplify and undefined as is.

Such forms are known as indeterminate forms. There are 7 types

$$\frac{0}{0},\ \frac{\infty}{\infty},\ 0 \cdot \infty,\ \infty - \infty,\ 0^0,\ \infty^0,\ 1^\infty$$

the are called indeterminate because as limits of $f(x)$ and $g(x)$ if, say, $\lim\limits_{x \to a} \frac{f(x)}{g(x)} \approx \frac{0}{0}$ then it could converge to any value.

To evaluate such limits, we often use L'Hôpital's Rule.

We will discuss $\frac{0}{0}$ and $\frac{\infty}{\infty}$ cases for now.

$\frac{0}{0}$ and $\frac{\infty}{\infty}$ cases

In general, if a limit is of the form

$$\lim_{x \to a} \frac{f(x)}{g(x)} \quad \text{with } \lim_{x\to a} f(x) = 0 \text{ and } \lim_{x \to a} g(x) = 0$$

we have some methods to solve but are still limited

Consider $\displaystyle\lim_{x \to 1} \frac{x^2 - x}{x^2 - 1} = \frac{1}{2}$

and $\displaystyle\lim_{x \to 0} \frac{\sin x}{x} = 1$

likewise for $\displaystyle\lim_{x\to a} \frac{f(x)}{g(x)}$ if $\displaystyle\lim_{x\to a} f(x) = \infty$

$\displaystyle\lim_{x \to a} g(x) = \infty$

Consider $\displaystyle\lim_{x \to \infty} \frac{x^2 - 1}{2x^2 + 1} = \frac{1}{2}$

but the usual methods don't work for

$$\lim_{x \to 1} \frac{\ln(x)}{x-1} \quad \text{or} \quad \lim_{x \to \infty} \frac{\ln(x)}{x-1}$$

How to evaluate?

We introduce a new approach

L'Hospital? V'', V

__Thm:__ L'Hôpital's Rule

- Suppose f and g are differentiable and $g'(x) \neq 0$ around a (except possibly at a).

  Suppose

  $$\lim_{x \to a} f(x) = 0 \quad \text{and} \quad \lim_{x \to a} g(x) = 0$$

  or

  $$\lim_{x \to a} f(x) = \pm\infty \quad \text{and} \quad \lim_{x \to a} g(x) = \pm\infty$$

  i.e. we have a $\frac{0}{0}$ or $\frac{\infty}{\infty}$ indeterminate form

  Then $$\lim_{x \to a} \frac{f(x)}{g(x)} = \lim_{x \to a} \frac{f'(x)}{g'(x)}$$

  given the latter limit exists or is infinite.

L'Hôpitals Rule is tedious to prove in general, but here is a specific case that is easier to see.

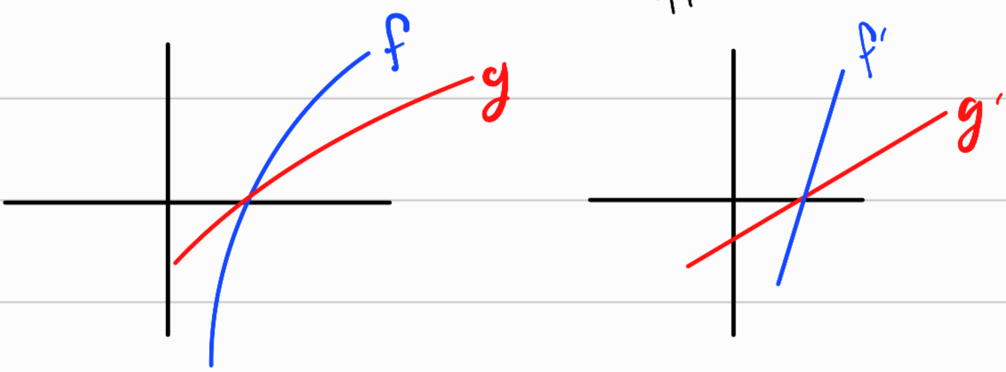
"Pf"/ If $f(a)=0$, $g(a)=0$, f' and g' continuous with $g'(a)\neq 0$

$$\lim_{x\to a}\frac{f'(x)}{g'(x)}=\frac{f'(a)}{g'(a)}=\frac{\lim\limits_{x\to a}\frac{f(x)-f(a)}{x-a}}{\lim\limits_{x\to a}\frac{g(x)-g(a)}{x-a}}$$

$$=\lim_{x\to a}\frac{f(x)-f(a)}{g(x)-g(a)}$$

$$=\lim_{x\to a}\frac{f(x)}{g(x)} \qquad \square$$

In the general case, we use the idea of linear approximators, that if you "zoom-in" to a smooth a function, it looks linear. Then the derivative is the best linear approximator



If they were linear, $f(x) \approx f'(a)(x-a)$

$g(x) \approx g'(a)(x-a)$

So $\dfrac{f(x)}{g(x)} \approx \dfrac{f'(a)(x-a)}{g'(a)(x-a)} = \dfrac{f'(a)}{g'(a)}$

So

$$\lim_{x \to a} \frac{f(x)}{g(x)} = \lim_{x \to a} \frac{f'(x)}{g'(x)}$$

Ex:

$$\lim_{x \to 1} \frac{\ln(x)}{x-1} \overset{L'H}{=} \lim_{x \to 1} \frac{1/x}{1} = \lim_{x \to 1} \frac{1}{x} = 1$$

$0 \cdot \infty$ Indeterminante case

If $\lim_{x \to a} f(x) = 0$, $\lim_{x \to a} g(x) = \pm\infty$, then

$\lim_{x \to a} f(x)g(x)$ is unknown

Consider, $\lim_{x \to 0^+} x^2 = 0$, $\lim_{x \to 0^+} \frac{1}{x} = \infty$

$$\lim_{x \to 0^+} x^2 \frac{1}{x} = 0$$

$$\lim_{x \to 0} x \frac{1}{x^2} = 0$$

$$\lim_{x \to 0^+} x \frac{1}{x} = 1$$

How do we deal with this?

Answer: Algebra.

If $\lim_{x \to a} f(x) = 0$, $\lim_{x \to a} g(x) = \pm\infty$, rewrite

$\lim_{x \to a} f(x)g(x)$ as $\lim_{x \to a} \frac{f(x)}{1/g(x)}$ or $\lim_{x \to a} \frac{g(x)}{1/f(x)}$

then $\lim_{x\to a} \frac{1}{g(x)} = \frac{1}{\pm\infty} = 0$, or $\lim_{x\to a} \frac{1}{f(x)} = \frac{1}{0} = \infty$

now a $\frac{0}{0}$ or $\frac{\infty}{\infty}$ case in which we use L'Hôpital's.

Ex: $\lim_{x\to 0^+} x \ln x$

$$= \lim_{x\to 0^+} \frac{\ln(x)}{1/x}, \quad \frac{\infty}{\infty} \text{ case}$$

$$\stackrel{L'H}{=} \lim_{x\to 0^+} \frac{1/x}{-1/x^2}$$

$$= \lim_{x\to 0} -x \; \boxed{=0}$$

Ex: What's the end behavior of $f(x) = xe^x$?

$$\lim_{x\to\infty} xe^x = \infty$$

$$\lim_{x\to-\infty} xe^x, \quad \infty\cdot 0 \text{ case}$$

$$= \lim_{x\to-\infty} \frac{x}{e^{-x}} \stackrel{L'H}{=} \lim_{x\to-\infty} \frac{1}{-e^{-x}} = \lim_{x\to-\infty} -e^x$$

$$= 0$$

$\infty - \infty$ Case

If $\lim_{x \to a} f(x) = \infty$ and $\lim_{x \to a} g(x) = \infty$

$\lim_{x \to a} f(x) - g(x)$ is unknown.

We want to force this into a $\frac{0}{0}$ or $\frac{\infty}{\infty}$ case.

How? Find common factor

Ex:

$$\lim_{x \to \frac{\pi}{2}^-} \sec(x) - \tan(x)$$

$$= \lim_{x \to \frac{\pi}{2}^-} \frac{1}{\cos(x)} - \frac{\sin(x)}{\cos(x)}$$

$$= \lim_{x \to \frac{\pi}{2}^-} \frac{1 - \sin(x)}{\cos(x)}, \quad \frac{0}{0} \text{ case}$$

$$\overset{L'H}{=} \lim_{x \to \frac{\pi}{2}^-} \frac{\cos(x)}{\sin(x)} = \lim_{x \to \pi/2} \cot(x) = 0$$

Ex: $\lim_{x\to\infty} e^x - x$

$= \lim_{x\to\infty} x\left(\frac{e^x}{x} - 1\right) = \infty$ since $\lim_{x\to\infty} \frac{e^x}{x} = \infty$ by L'H

0^0, ∞^0, 1^∞ Cases

In these cases we use properties of e^x and $\ln x$.

If $\lim_{x\to a} f(x)^{g(x)}$ results in a 0^∞, ∞^0, or 1^∞ case, we do the following

$$\lim_{x\to a} f(x)^{g(x)} = \lim_{x\to a} e^{\ln\left(f(x)^{g(x)}\right)}$$

$$= \lim_{x\to a} e^{g(x)\ln(f(x))}$$

$$= e^{\lim_{x\to a} g(x)\ln(f(x))}$$

(This step is allowed as e^x is continuous

we now just need to evaluate
$\lim_{x\to a} g(x)\ln(f(x))$ which we already know how
to do.

Ex: $\lim_{x\to 0^+} x^x$ a 0^0 case.

$= \lim_{x\to 0^+} e^{x\ln x}$

$= e^{\lim_{x\to 0^+} x\ln x}$ ⤳ $\lim_{x\to 0^+} x\ln x$ a $0\cdot\infty$ case

$= \lim_{x\to 0^+} \frac{\ln(x)}{1/x}$ a $\frac{\infty}{\infty}$ case

$\overset{L'H}{=} \lim_{x\to 0^+} \frac{1/x}{-1/x^2}$

$= \lim_{x\to 0^+} -x = 0$

$= e^0$

$$\boxed{\lim_{x\to 0^+} x^x = 1}$$

Ex:

$$\lim_{x\to 0^+}\left(1+\sin(4x)\right)^{\cot(x)}$$ a 1^∞ case

$$=\lim_{x\to 0^+} e^{\cot(x)\ln(1+\sin 4x)}$$

$$\lim_{x\to 0^+}\cot x\,\ln\left(1+\sin 4x\right)$$ $\infty\cdot 0$ case

$$=\lim_{x\to 0^+}\frac{\ln(1+\sin 4x)}{1/\cot x}$$

$$=\lim_{x\to 0^+}\frac{\ln(1+\sin 4x)}{\tan x}$$, $\frac{0}{0}$ case

$$\stackrel{L'H}{=}\lim_{x\to 0^+}\frac{\frac{4\cos(4x)}{1+\sin(4x)}}{\sec^2 x}$$

$$=\frac{\frac{4(1)}{1+0}}{1^2}=4$$

$$\boxed{=e^4}$$